Ahmad Jawed Rahimi

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EDUCATION

University of California, Berkeley **Berkeley, CA**
B.S. Electrical Engineering and Computer Sciences *Expected Dec 2026*

GPA: 3.7/4.0 | **Honors:** College of Engineering Dean's List, Fall 2025 and Spring 2026

Relevant Coursework: Microelectronic Devices & Circuits, Microfabrication Technology, Integrated-Circuit Devices, Digital Design & Integrated Circuits, FPGA Laboratory, Optical Engineering, Electric Power & Renewable Energy, Circuits & Devices, Computer Architecture

ENGINEERING EXPERIENCE

NASA L'SPACE Program **Remote**
Engineering Lead / Electrical Systems Engineer - NPWEE *Sep 2026 - Present*

- Lead development of a space-system fault-detection concept using current and voltage sensing, automated branch monitoring, and electrical isolation to identify degraded or failed power paths.
- Evaluate sensing, switching, embedded-control, and prototype architectures while supporting requirements analysis, risk assessment, technical feasibility, and NASA-style documentation.

Electrical Engineer - Mission Concept Academy *Jun 2026 - Sep 2026*

- Supported electrical and power-subsystem design for a multidisciplinary lunar rover mission with constrained mass, power, cost, and operational requirements.
- Developed electrical requirements, subsystem architecture, power-budget assumptions, verification plans, procurement considerations, and fault-recovery strategies for Preliminary Design Review documentation.
- Analyzed solar generation, lithium-ion battery storage, PDU distribution, BMS monitoring, safe-power modes, and payload, mobility, thermal, and CDH loads; collaborated across subsystem teams to define interfaces and identify integration risks.

Berkeley Vertical Farming **Berkeley, CA**
Electrical Engineer *Aug 2025 - Present*

- Design, assemble, and test automation and sensor-monitoring systems for hydroponic applications, integrating electrical hardware, control software, and operational requirements.
- Troubleshoot sensor, wiring, and hardware/software integration issues; prototype nutrient and water-delivery mechanisms with an interdisciplinary engineering and plant-science team.

UC Berkeley Electrical Engineering and Computer Sciences **Berkeley, CA**
Undergraduate Researcher *Jun 2025 - May 2026*

- Developed automated real-time test pipelines for Lingua Franca federated programs, integrating Raspberry Pi hardware runners into GitHub Actions to evaluate timing under real network and clock variability.
- Implemented diagnostics for deadline misses, clock drift, and nondeterministic behavior, supporting systematic failure analysis and validation of distributed embedded systems.

UC Berkeley Formula Racing **Berkeley, CA**
Electrical Engineer - Harness *Aug 2024 - Dec 2025*

- Constructed and validated a high-reliability vehicle electrical harness, performing connection inspection, continuity testing, and systematic troubleshooting to maintain a failure rate below 1%.

HARDWARE & LABORATORY PROJECTS

Microelectronic Circuit Testing | *EE 105*

- Built and debugged transistor- and op-amp-based circuits from schematics; used oscilloscopes, DMMs, function generators, and bench power supplies to verify rails, trace signals, measure gain and frequency response, and isolate circuit faults.

FPGA Digital Systems | *SystemVerilog, Vivado, AUP-ZU3*

- Implemented and tested digital hardware on a Xilinx Zynq UltraScale+ FPGA; used simulation, waveform analysis, hardware programming, and board-level testing to diagnose logic and interface behavior.

Treble Boost Effect Box | *Analog Circuit Design*

- Designed and assembled an op-amp active high-pass filter; calculated component values and used bench instruments to compare input/output waveforms, measure frequency response, trace the signal path, and tune gain.

TECHNICAL SKILLS

Bench Test & Debug: Oscilloscopes, digital multimeters, function generators, bench power supplies, continuity testing, voltage measurement, waveform analysis, frequency-response measurement, signal tracing, power-rail verification, and schematic reading.
Hardware & Assembly: Electrical harness construction, connector and wiring inspection, sensor integration, circuit assembly, Arduino, Raspberry Pi, op-amp circuits, and active filters
Microelectronics: Semiconductor devices, transistor circuits, IC-device fundamentals, microfabrication processes, cleanroom procedures, contamination control, and ESD-aware component handling
Digital & Embedded: SystemVerilog, FPGA development, digital logic, UART, FIFOs, ready-valid interfaces, computer architecture, RISC-V, C/C++, Python, and Java
Tools: Vivado, Logisim Evolution, MATLAB, AutoCAD, SolidWorks, Git/GitHub, GitHub Actions, Linux/Unix, and VS Code